PS1 and PS2 Helper Slides

Optimization TA board session

Kacper Krasowski

TA Sessions — Logistics

Who am I?

Kacper Krasowski

Research interests:
Digital Economics: Social Media, AI, LLMs

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Office hours:
Just send me an email — I am here basically every day.
Maximum: 1 hour per week for all of you.

How the TA works

- I will be showing you the solutions to the Problem Sets. (Hopefully correct)
- Roughly every second week: a short test to check your understanding.
- Problem sets are not graded, but they are definitely worth doing.

Problem Set 1

Basic Set Theory, Ordered Sets, Metric Spaces



PS1 Exercise 1

Exercise statement

Problem sheet text

Prove

$$(S \cap T)^c = S^c \cup T^c.$$

PS1 Exercise 1

Prove $(S \cap T)^c = S^c \cup T^c$.

Complement in X

$$S^c = \{x \in X : x \notin S\} = X \setminus S.$$

Set operations

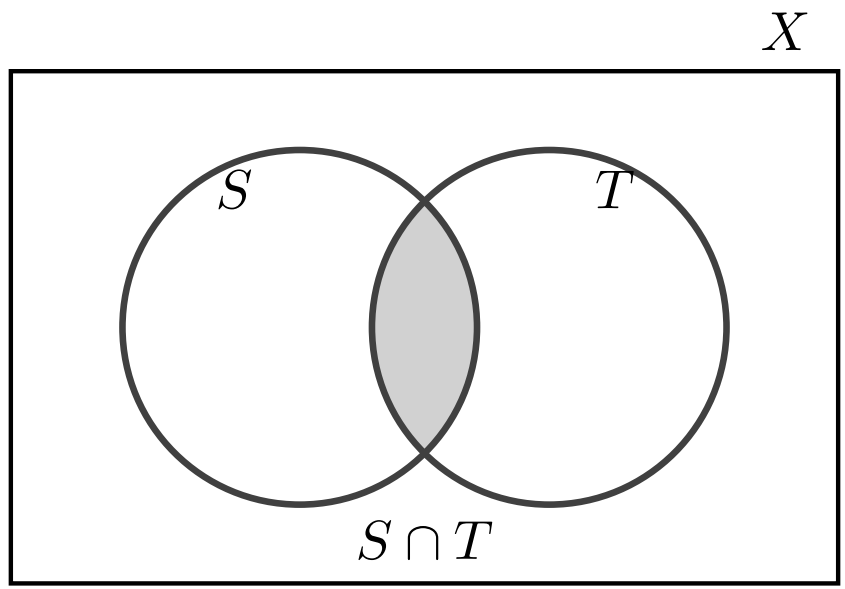
$$S \cup T = \{x : x \in S \text{ or } x \in T\},$$
$$S \cap T = \{x : x \in S \text{ and } x \in T\}.$$

PS1 Exercise 1

Venn intuition for $(S \cap T)^c = S^c \cup T^c$.

Proof steps

$$x \notin S \cap T$$

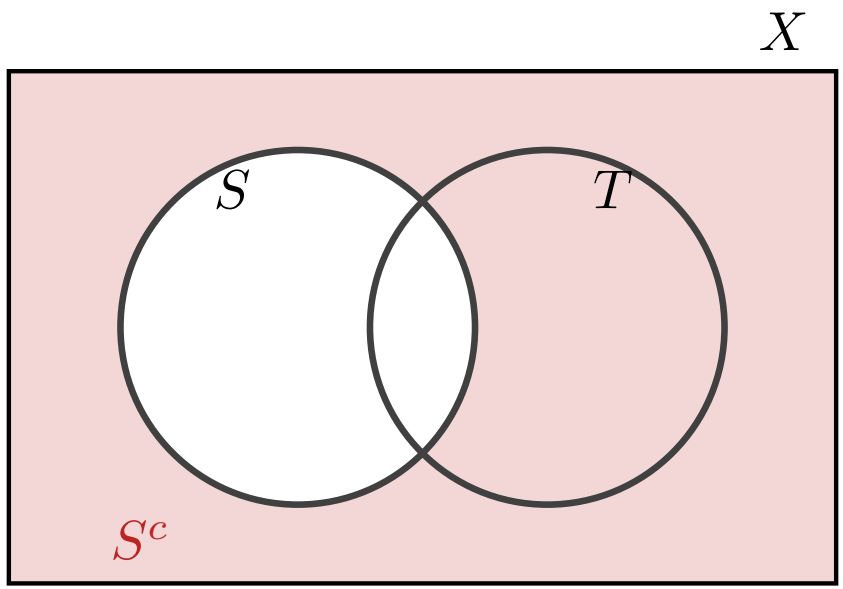


PS1 Exercise 1

Venn intuition for $(S \cap T)^c = S^c \cup T^c$.

Proof steps

$$x \notin S \cap T \Rightarrow x \notin S \text{ or } x \notin T$$

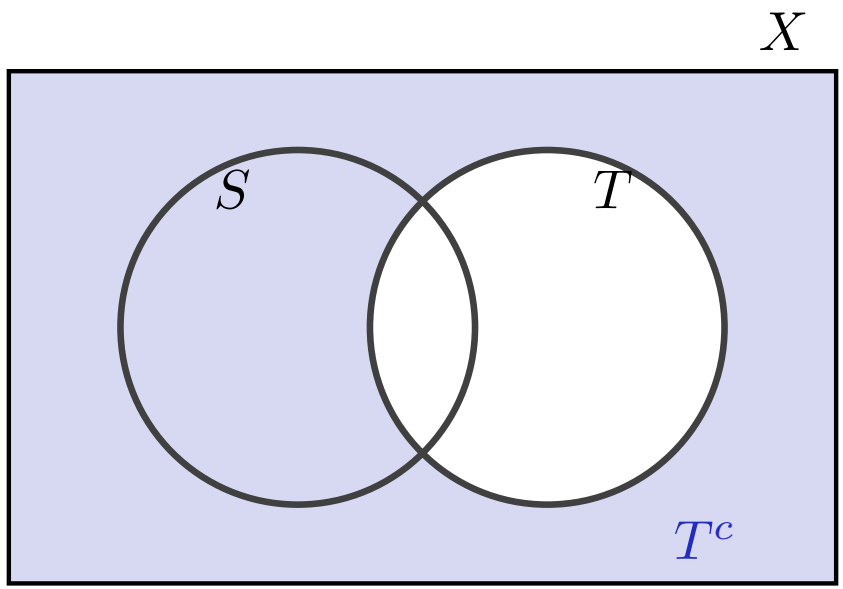


PS1 Exercise 1

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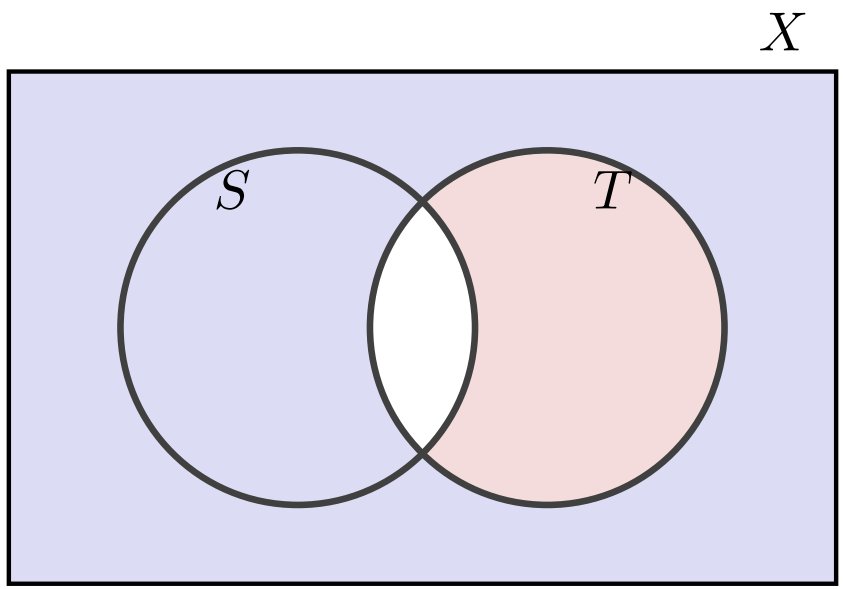


PS1 Exercise 1

Venn intuition for $(S \cap T)^c = S^c \cup T^c$.

Proof steps

$$x \notin S \cap T \Rightarrow x \in S^c \text{ or } x \in T^c \Rightarrow x \in S^c \cup T^c$$



$S^c \cup T^c$

PS1 Exercise 2

Exercise statement

Problem sheet text

Show that any equivalence relation on a set X partitions X.
(Exercise 1.17 in Carter.)

PS1 Exercise 2

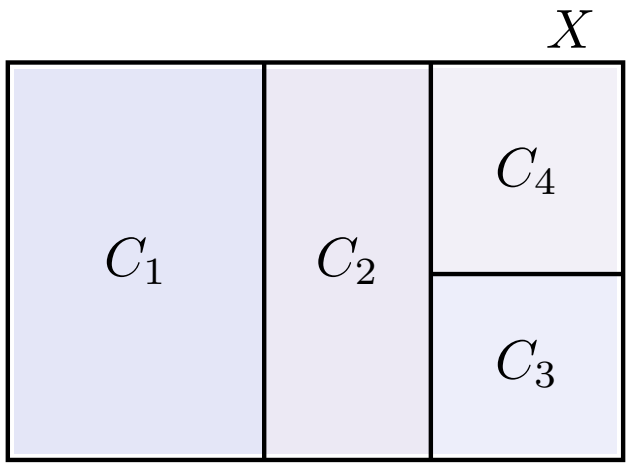
What does "partitions X" mean?

Partition of X

A collection of subsets is a partition of X when:

1. the subsets together cover X;
2. two different members do not share an element.

Equivalently: distinct subsets are disjoint.



PS1 Exercise 2

Show that any equivalence relation on a set X partitions X.

Equivalence relation

A relation $\sim$ on X is an equivalence relation if it is:

$$\begin{aligned} &\text{reflexive:} && x \sim x, \\ &\text{symmetric:} && x \sim y \Rightarrow y \sim x, \\ &\text{transitive:} && x \sim y,\ y \sim z \Rightarrow x \sim z. \end{aligned}$$

Equivalence class

For $a \in X$,

$$\sim(a) = \{x \in X : a \sim x\}.$$

PS1 Exercise 2

Different equivalence classes are disjoint.

Proof tool: contrapositive

$$P \Rightarrow Q \quad \Longleftrightarrow \quad \neg Q \Rightarrow \neg P.$$

This is not the same as reversing the implication $Q \Rightarrow P$.

PS1 Exercise 3

Exercise statement

Problem sheet text

Consider an ordered set. Show that every best element must be a maximal element. Show that a maximal element need not be a best element. (Exercise 1.27 in Carter.)

PS1 Exercise 3

Every best element is maximal, but not vice versa.

Strict dominance

$$x \succ y \quad \Longleftrightarrow \quad x \succeq y \text{ and not } y \succeq x.$$

Maximal and best elements

$$x \in X \text{ is maximal} \Longleftrightarrow \nexists z \in X \text{ such that } z \succ x,$$
$$x \in X \text{ is best} \Longleftrightarrow \forall z \in X,\ x \succeq z.$$

Distinction

BEST: dominates every element,

MAXIMAL: is not strictly dominated by any element.

PS1 Exercise 3

Why maximal need not mean best.

Disproving a universal implication

$$P: \ x \text{ is maximal} \qquad\qquad Q: \ x \text{ is best}$$

To disprove

"every maximal element is best,"

find one ordered set and one $x \in X$ with

$$P \text{ true} \qquad \text{and} \qquad Q \text{ false.}$$

PS1 Exercise 4

Exercise statement

Problem sheet text

Show that every finite ordered set has at least one maximal element. (Exercise 1.28 in Carter.)

PS1 Exercise 4

Show that every finite ordered set has at least one maximal element.

Finite set

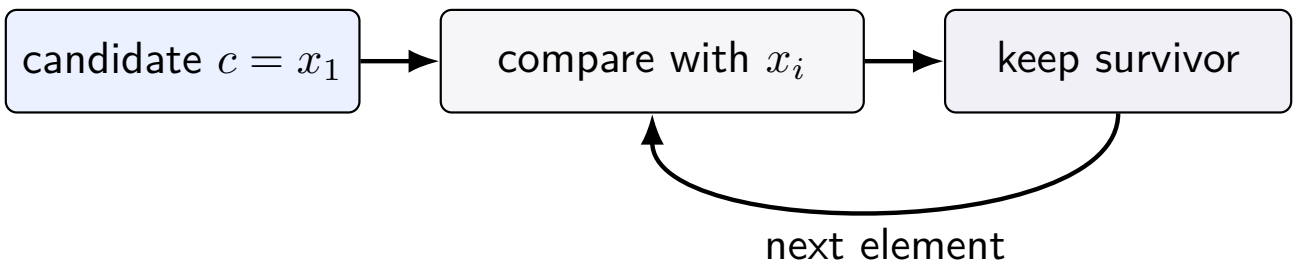
$$X = \{x_1, x_2, \ldots, x_n\}.$$

Maximal

$$\nexists z \in X \quad z \succ x.$$

PS1 Exercise 4

Board-process reminder.



PS1 Exercise 5

Exercise statement

Problem sheet text

Show that maximal and best elements coincide in any weakly ordered set. That is,

$$x \text{ is maximal} \iff x \text{ is best.}$$

(Exercise 1.54 in Carter. Weak orders are complete and transitive.)

PS1 Exercise 5

Maximal and best elements coincide in any weakly ordered set.

Order definitions

$$x \text{ maximal} \iff \nexists z \in X : \ z \succ x,$$
$$x \text{ best} \iff \forall z \in X, \ x \succeq z.$$

Weak order ingredients

Complete:

$$\forall x, z \in X, \quad x \succeq z \text{ or } z \succeq x.$$

Transitive:

$$x \succeq y, \ y \succeq z \Rightarrow x \succeq z.$$

PS1 Exercise 6

Exercise statement

Problem sheet text

Show that open balls are open sets.

PS1 Exercise 6

Show that open balls are open sets.

Open ball

$$B_r(x_0) = \{x \in X : \rho(x, x_0) < r\}.$$

Triangle inequality

$$\rho(x, z) \leq \rho(x, y) + \rho(y, z).$$

Open set

$S \subseteq X$ is open if for every $y \in S$ there exists $\epsilon > 0$ such that

$$B_\epsilon(y) \subseteq S.$$

PS1 Exercise 7

Exercise statement

Problem sheet text

Show that Definitions 2 and 4 are equivalent.

PS1 Exercise 7

Show that Definitions 2 and 4 are equivalent.

Definition 2 (Open set)

Given any metric space $\langle X, \rho \rangle$, a subset $S \subseteq X$ is said to be open, if for any $x_0 \in S$ there exists $r > 0$ such that $B_r(x_0) \subseteq S$.

Definition 3 (Neighborhood and interior point)

Given any metric space $\langle X, \rho \rangle$ and any point $x_0 \in X$, a subset $S \subseteq X$, such that $x_0 \in S$, is said to be a neighborhood of x_0 if there exists $r > 0$ such that $B_r(x_0) \subseteq S$. If S is a neighborhood of x_0 then x_0 is an interior point of S.

Definition 4 (Open set)

Given any metric space $\langle X, \rho \rangle$, a subset $S \subseteq X$ is said to be open, if it is a neighborhood of all its points.

PS1 Exercise 7

Show that Definitions 2 and 4 are equivalent.

Definition 2 (Open set)

Given any metric space $\langle X, \rho \rangle$, a subset $S \subseteq X$ is said to be open, if for any $x_0 \in S$ there exists $r > 0$ such that $B_r(x_0) \subseteq S$.

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Definition 4 (Open set)

Given any metric space $\langle X, \rho \rangle$, a subset $S \subseteq X$ is said to be open, if it is a neighborhood of all its points.

PS1 Exercise 8

Exercise statement

Problem sheet text

Provide an example of a nonempty subset of $\mathbb{R}$ which is not open under the standard metric

$$\rho(x, y) = |x - y|,$$

and show that it is not open.

PS1 Exercise 8

Provide a nonempty subset of $\mathbb{R}$ which is not open under $\rho(x, y) = |x - y|$.

Open

$$\forall x \in S \; \exists r > 0 : B_r(x) \subseteq S.$$

Not open

$$\exists x \in S \; \forall r > 0 : B_r(x) \not\subseteq S.$$

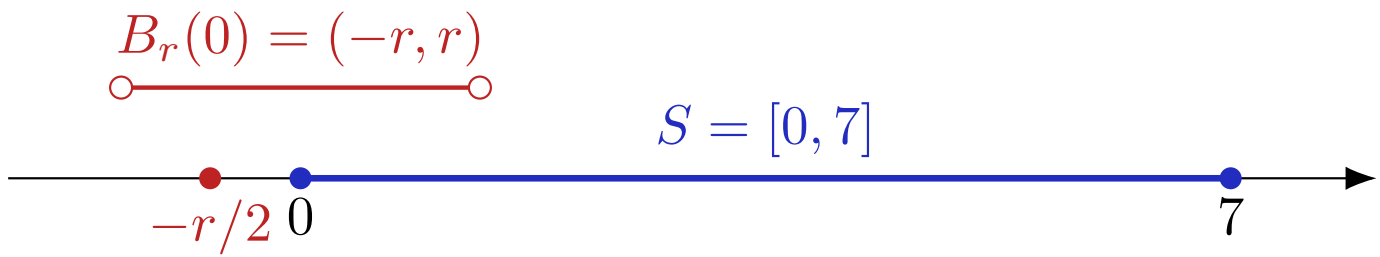
$\forall x \; \exists r$ → $\exists x \; \forall r$

Criterion

To show a set is not open, find one bad point in the set such that every ball around it contains a point outside the set.

PS1 Exercise 8

Use the example $S = [0, 7]$.



Visual cue

Every interval around the endpoint 0 leaks outside $[0, 7]$.

PS1 Exercise 9

Exercise statement

Problem sheet text

Prove that any set of the form

$$\{(x, x_0) \in \mathbb{R}^2 : a < x < b\},$$

where $x_0 \in \mathbb{R}$ is fixed, is not open in $\mathbb{R}^2$ under the standard Euclidean metric.

PS1 Exercise 9

Prove $\{(x, x_0) \in \mathbb{R}^2 : a < x < b\}$ is not open in $\mathbb{R}^2$ under the Euclidean metric.

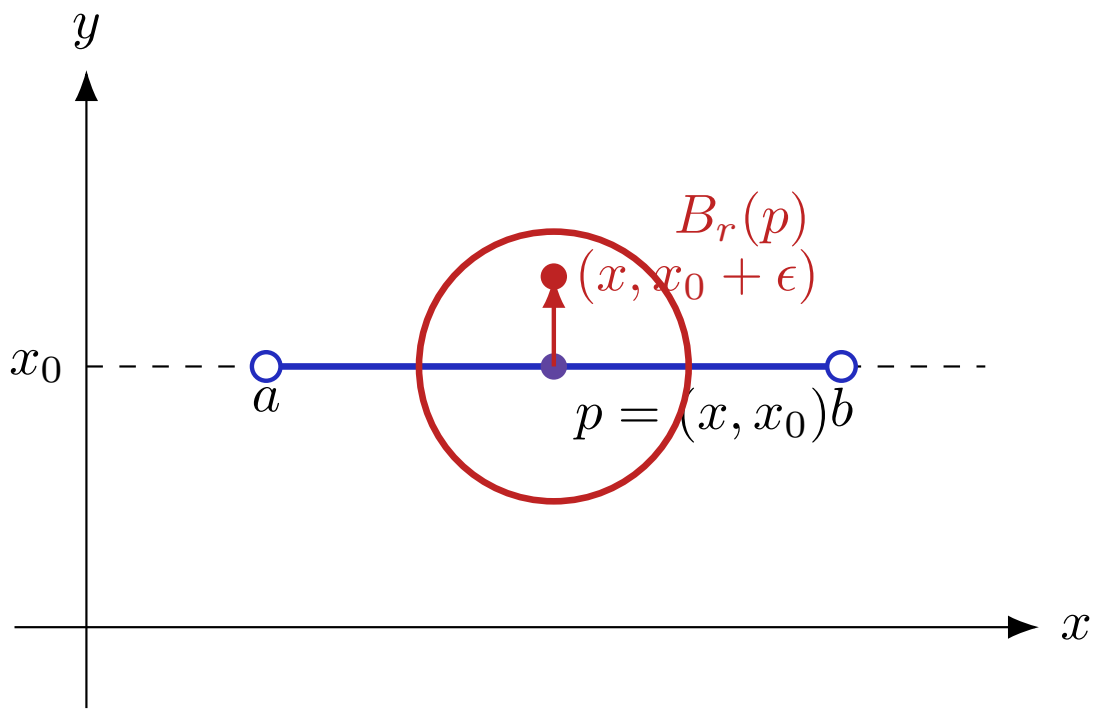
Same non-openness criterion

Find one point $p \in S$ such that every Euclidean ball around p contains a point outside S.

$$S = \{(x, x_0) \in \mathbb{R}^2 : a < x < b\}.$$

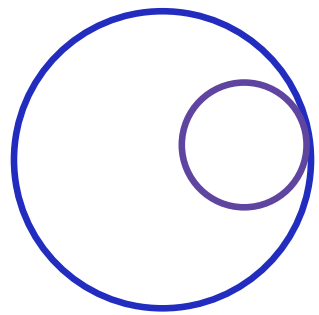
PS1 Exercise 9

A line segment has no two-dimensional room.



The vertical perturbation stays in the ball but leaves the horizontal set.

Metric Spaces



open sets, closure, equivalent metrics, compactness

PS2 Exercise 1

Exercise statement

Problem sheet text

Show that given any metric space $\langle X, \rho \rangle$:

(a) The empty set and the entire set X are open.

(b) Given an arbitrary collection of open sets $\mathcal{C}$,

$$\bigcup_{S \in \mathcal{C}} S$$

is open.

(c) Given a finite collection of open sets $\mathcal{C}$,

$$\bigcap_{S \in \mathcal{C}} S$$

is open.

(d) Consider $\langle X, \rho \rangle$ with $X = \mathbb{R}$ and $\rho(x, y) = |x - y|$. Provide an example of an infinite collection $\mathcal{C}$ of open sets in $\mathbb{R}$ whose intersection is $[0, 1] \subseteq X$.

PS2 Exercise 1

Part (c): finite intersections of open sets.

Finite collection

A finite collection has the form

$$\mathcal{C} = \{S_1, S_2, \ldots, S_n\}.$$

PS2 Exercise 1

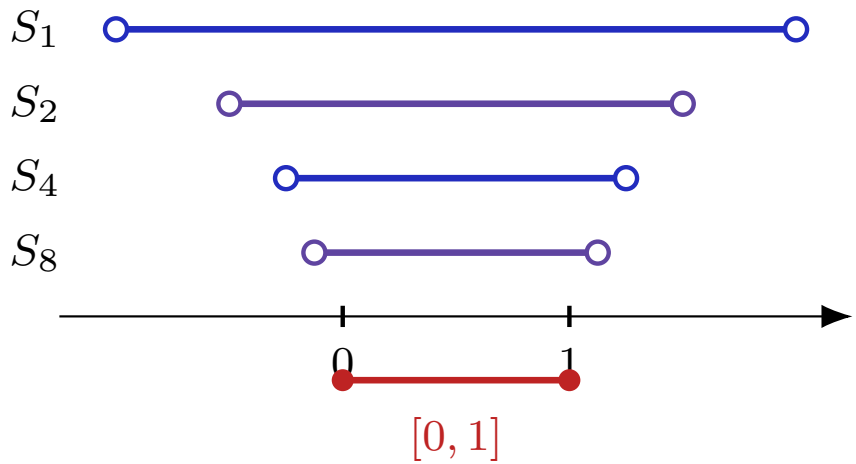
Part (d): infinite intersections can lose openness.

What goes wrong

Positive radii may shrink toward 0:

$$1,\ \frac{1}{2},\ \frac{1}{3},\ \frac{1}{4},\dots$$

There need not be a positive minimum.



Handwritten example

$$S_n = \left(-\frac{1}{n},\ 1+\frac{1}{n}\right)$$

$$\bigcap_{n=1}^{\infty} S_n = [0,1].$$

PS2 Exercise 2

Exercise statement

Problem sheet text

Let $\bar{S}$ be the closure of $S \subseteq X$. Show that $\bar{S}$ is the smallest closed subset of X containing S: for any closed set $T \subseteq X$, prove that

$$S \subseteq T \Rightarrow \bar{S} \subseteq T.$$

PS2 Exercise 2

The closure $\bar{S}$ is the smallest closed subset containing S.

Boundary point

$x \in X$ is a boundary point of $S \subseteq X$ if every open ball around x contains points of S and points of S^c.

Boundary and closure

$$\mathrm{bn}(S) = \{\text{boundary points of } S\},$$

$$\bar{S} = \mathrm{cl}(S) = S \cup \mathrm{bn}(S).$$

Closed set

A subset $T \subseteq X$ is closed if it contains all of its limit points.

PS2 Exercise 3

Exercise statement

Problem sheet text

Consider the Euclidean and sup metrics in $\mathbb{R}^N$:

$$\rho_2(x,y) = \sqrt{\sum_{i=1}^{N}(x_i - y_i)^2}, \qquad \rho_\infty(x,y) = \max_{i=1,\ldots,N} |x_i - y_i|.$$

Show that $S \subseteq \mathbb{R}^N$ is open under ρ_2 if and only if it is open under ρ_∞.

Hint: Given any open ball in one metric, find an open ball in the other metric contained in the first ball.

What about closedness, boundedness?

PS2 Exercise 3

Euclidean metric versus sup metric.

Metrics on $\mathbb{R}^N$

$$\rho_2(x,y) = \sqrt{\sum_{i=1}^{N}(x_i - y_i)^2}$$

$$\rho_\infty(x,y) = \max_{i=1,\ldots,N} |x_i - y_i|.$$

Handwritten route

Compare:

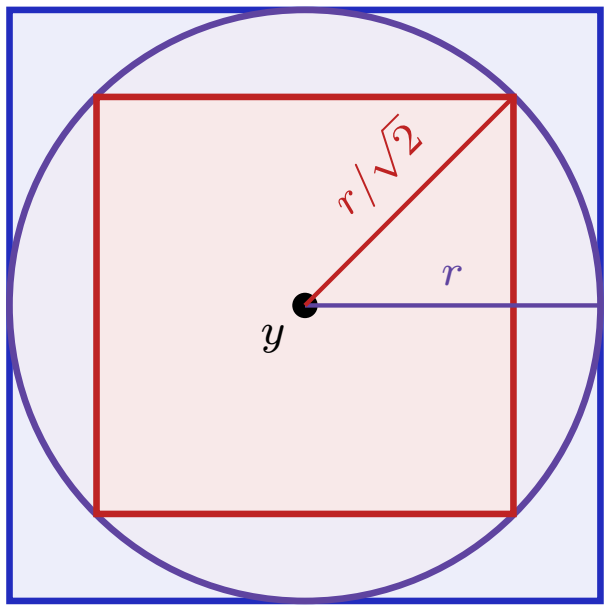
$$\max_i |x_i - y_i|^2 \quad \text{and} \quad \sum_{i=1}^{N}(x_i - y_i)^2.$$

Metric comparison

$$\rho_\infty(x,y) \leq \rho_2(x,y) \leq \sqrt{N}\,\rho_\infty(x,y).$$

PS2 Exercise 3

Square in circle in square.



$B_r^\infty(y)$

$B_r^2(y)$

$B_{r/\sqrt{2}}^\infty(y)$

In $\mathbb{R}^N$

$$B_{r/\sqrt{N}}^\infty(y) \subseteq B_r^2(y),$$
$$B_r^2(y) \subseteq B_r^\infty(y).$$

Also:

$$B_r^\infty(y) \subseteq B_{\sqrt{N}r}^2(y).$$

Proof idea

To transfer openness, put a ball from one metric inside the ball given by the other metric.

PS2 Exercise 3

Closedness and boundedness.

Closed sets

Since the open sets coincide under ρ_2 and ρ_∞, closed sets coincide by complements.

Bounded sets

The same inequalities compare bounded balls:

$$S \subseteq B_R^2(y) \Rightarrow S \subseteq B_R^\infty(y), \qquad S \subseteq B_R^\infty(y) \Rightarrow S \subseteq B_{\sqrt{N}R}^2(y).$$

PS2 Exercise 4

Exercise statement

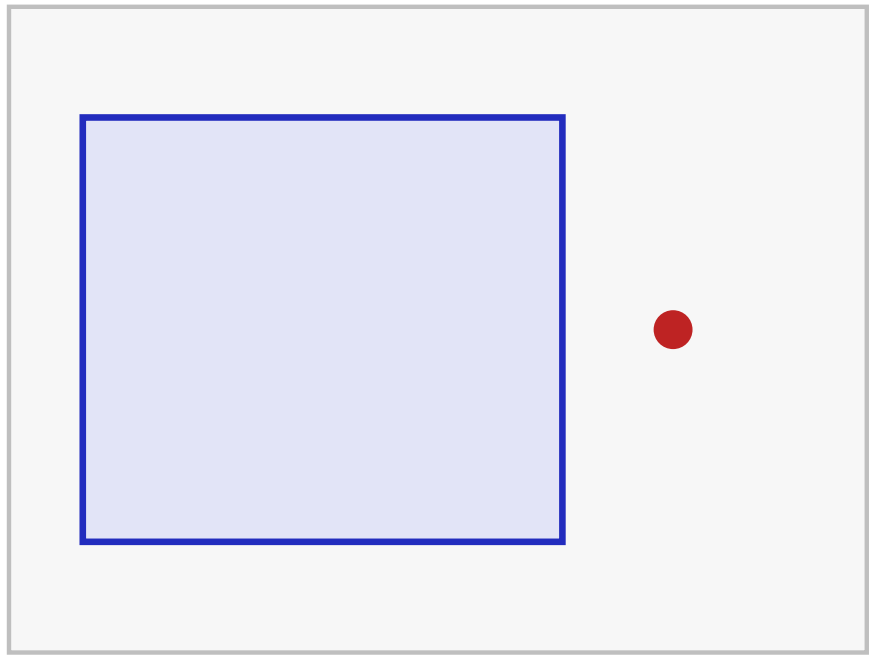
Problem sheet text

Consider $\langle X, \rho \rangle$ and let $S \subseteq X$ be a compact subset of X. Show that S is closed.

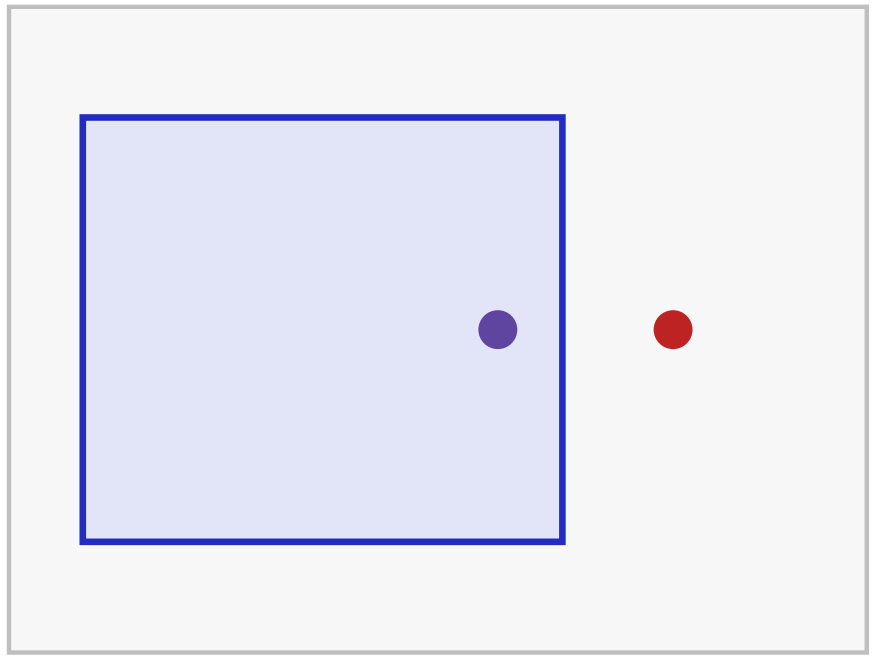
PS2 Exercise 4



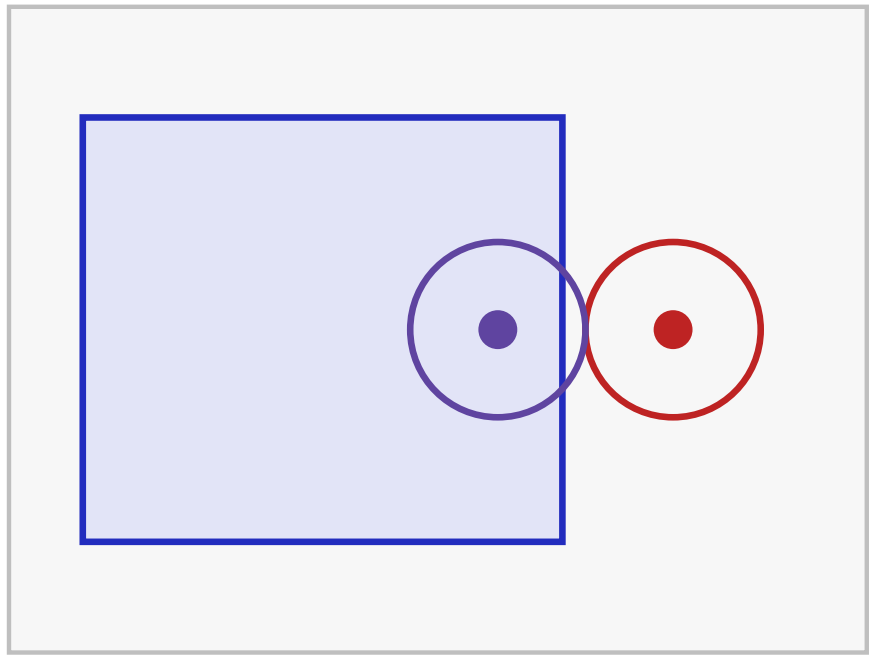
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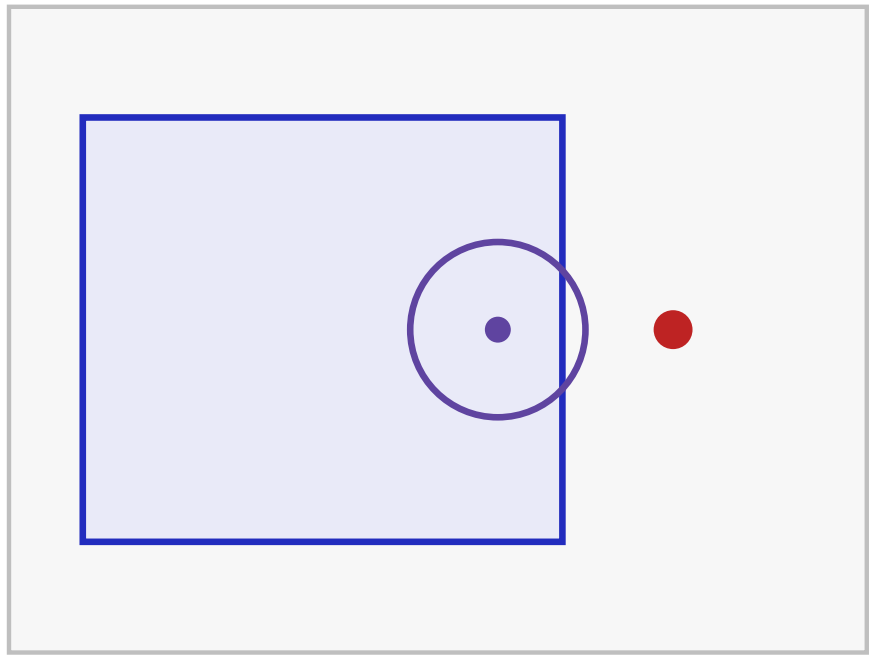
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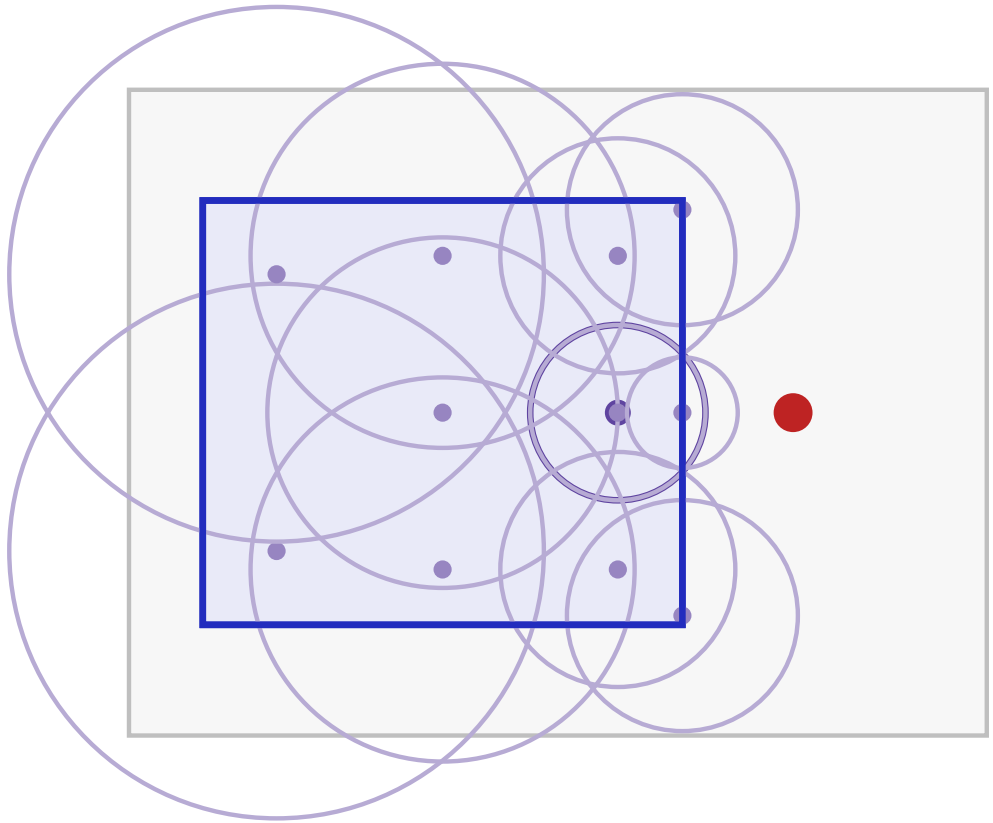
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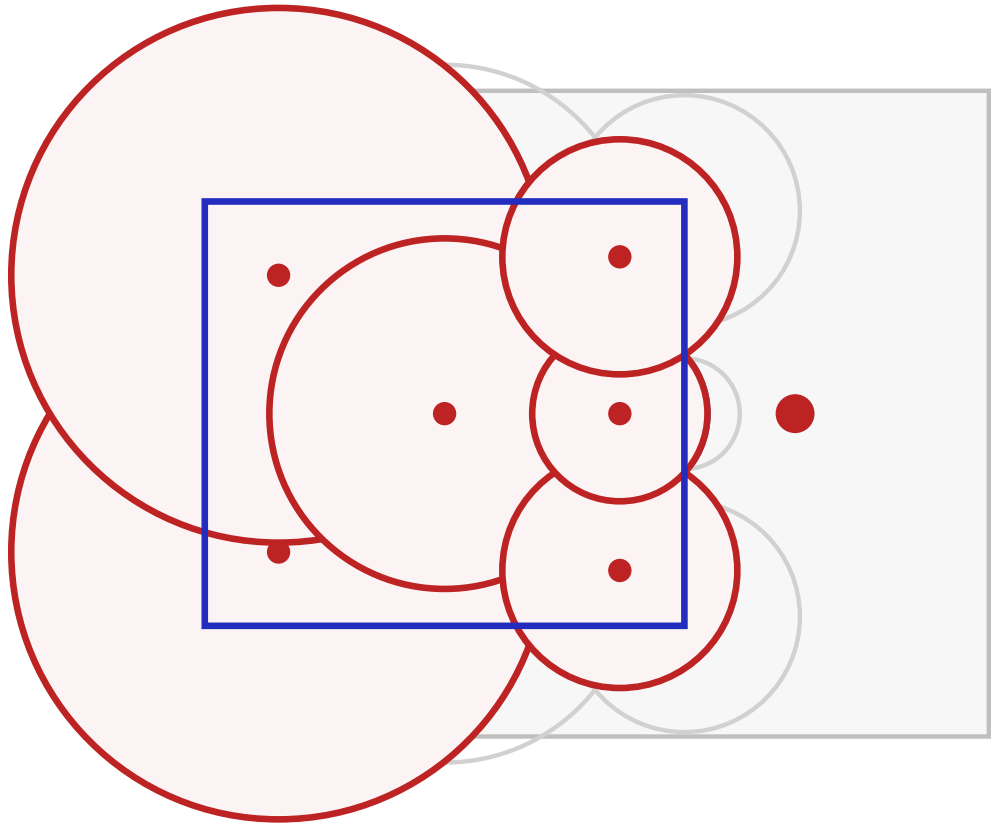
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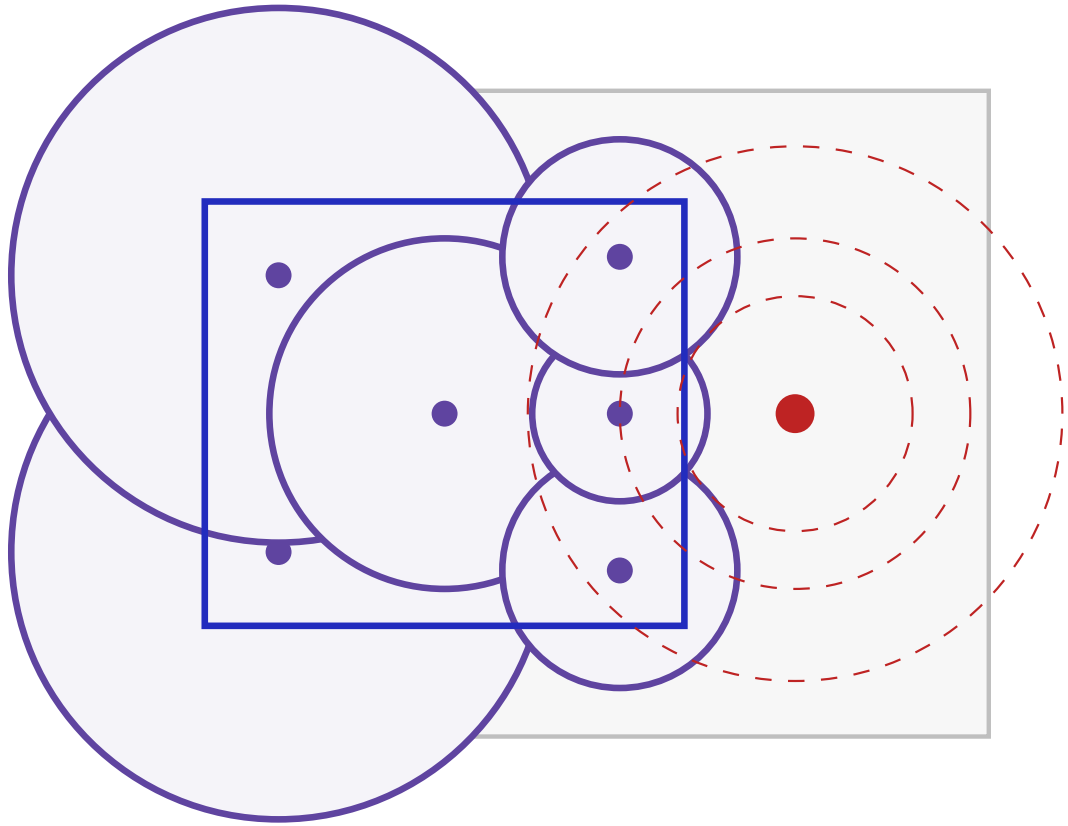
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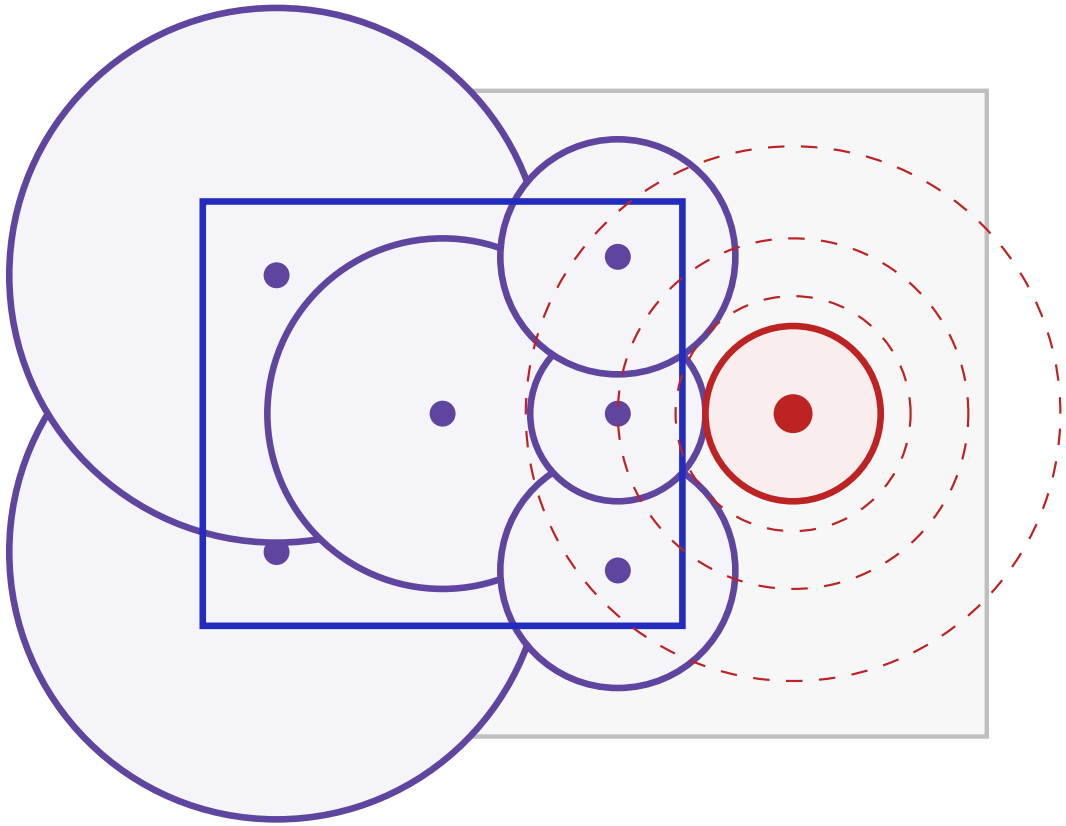
PS2 Exercise 4



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